VII. (35 points) 1. Nuclear electric force makes an electron circle in classical hydrogen. Classical EM theory says accelerated electrons emit waves; the radiated power is given by (Larmor's formula) $P=\frac{e^2a^2}{6\pi c^3\varepsilon_0}$; a is acceleration, c vacuum light speed, $\varepsilon_0=\frac{1}{4\pi k}=8.854\times10^{-12}\,\mathrm{Fm^{-1}}$; the electron charge magnitude is $e=1.602\times10^{-19}\,\mathrm{C}$. Neglect relativity and estimate a classical atom's lifetime τ. (Measured binding energy is E_B=13.6eV, and electron rest mass is m_0=9.109×10^{-31}kg）

2. Radiation from accelerated charges has important uses. Synchrotron radiation, widely used in modern research, comes from near-light-speed electrons curving in magnetic fields. Its core is the electron storage ring; bunches traverse bending magnets and other devices to make high-quality synchrotron light. The recently built Shanghai facility is a third-generation source, with electron energy E=3.50GeV and bunch current I=300mA, circumference L=432m, cells (bending-magnet turns) N=20, and dipole field B=1.27T. Find mean total radiated power per outlet P_O.

(Near-light-speed electrons obey Newton's second law if relativistic mass is constant; the Larmor formula becomes $P=\frac{e^2a^2}{6\pi c^3\varepsilon_0}\times\gamma^4$, where $\gamma=\frac{E}{m_0c^2}$, E is total energy, and m_0c^2 is electron rest energy.)

3. Near-light-speed ring electrons emit synchrotron radiation in a cone along the orbit tangent, with half-angle $1/\gamma$. Thus higher electron energy gives better directionality.

For this facility, find the duration of a single electron's radiation received at a point along the radiation direction: ΔT.

(Give all results in this problem to three significant figures.)